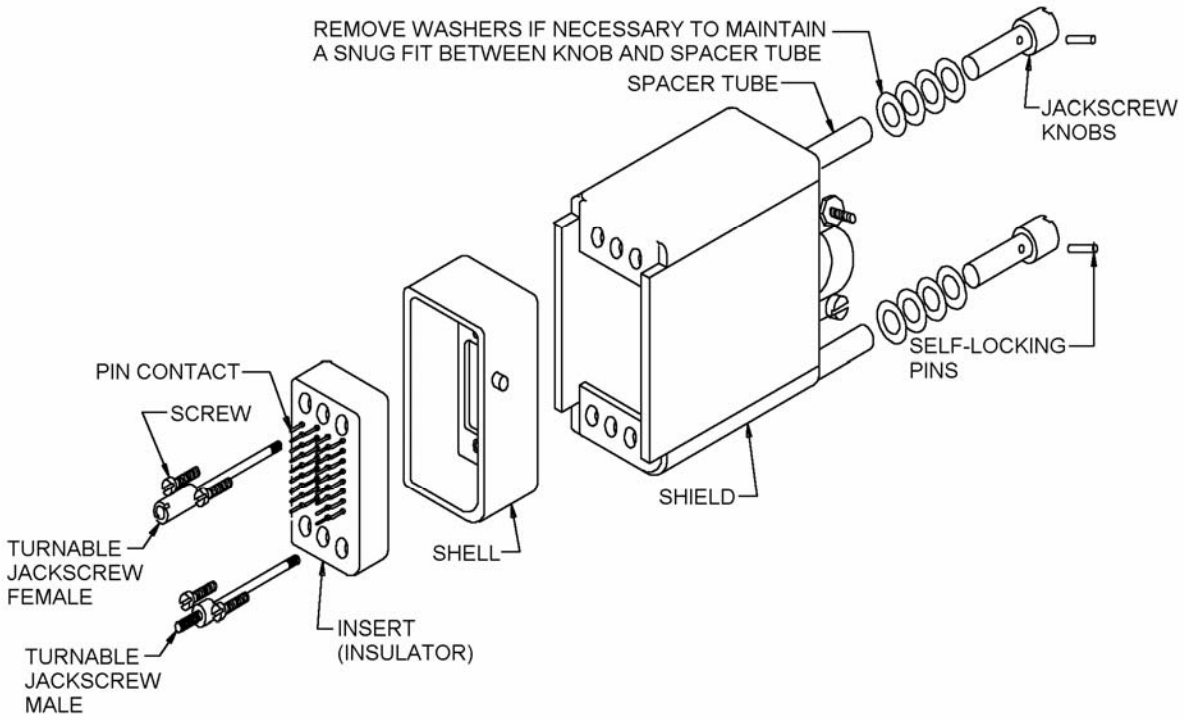
DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, HARDWARE FOR SIZES 16 AND 20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

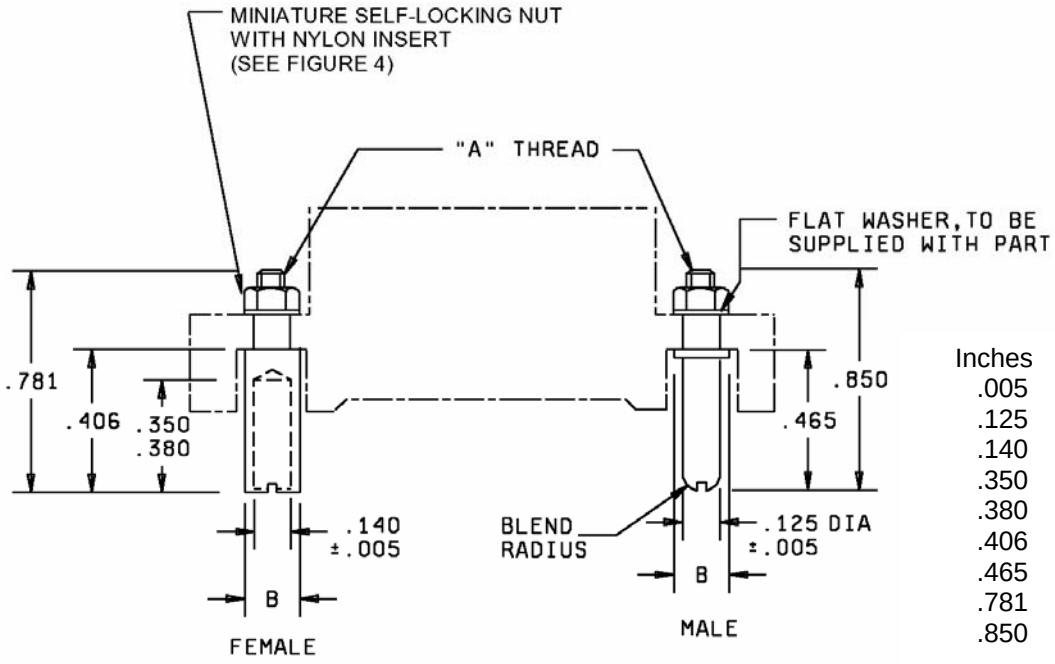
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.



Connector assembly

FIGURE 1. Plug connector with shield, plug shell, and turnable jackscrews.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Inches	mm
.005	0.13
.125	3.18
.140	3.56
.350	8.89
.380	9.65
.406	10.31
.465	11.81
.781	19.84
.850	21.59

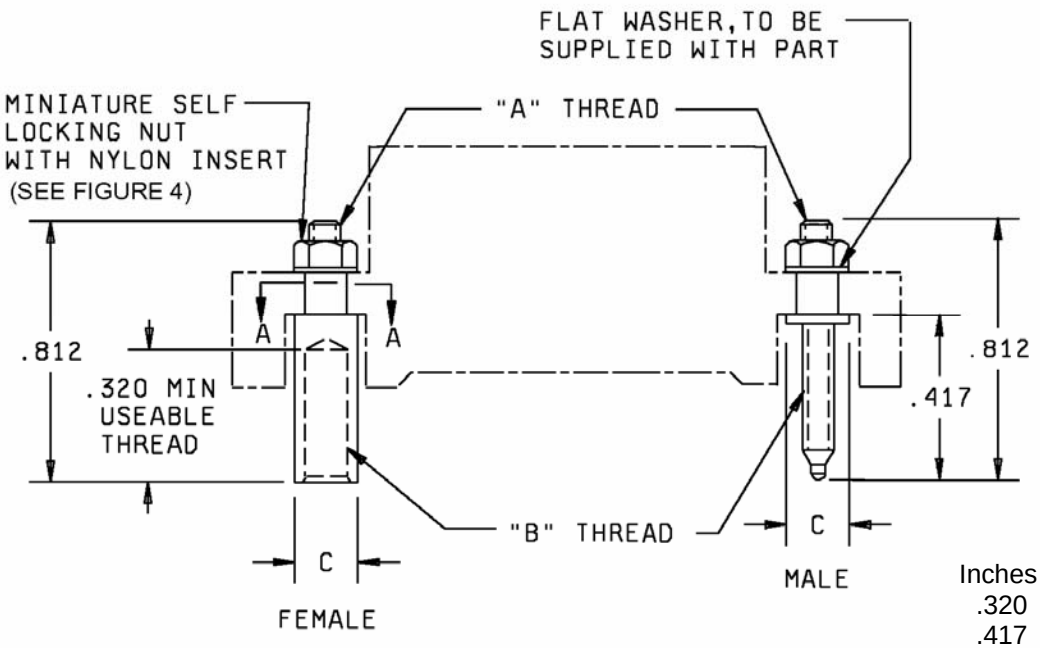
Guidepin male and female

Guide pin or guide socket designator	Gender	Connector size (number of contacts)	A Thread class 2	B dia max
M18197-1	Male	7, 9, 14, 18 20, 26, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	.190 (4.83)
M18197-2	Female	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	.190 (4.83)
M18197-3	Male	104	6-32 UNC-2A .138-32 UNC	.190 (4.83)
M18197-4	Female	104	6-32 UNC-2A .138-32 UNC	.190 (4.83)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material corrosion resistance steel in accordance with SAE-AMS-QQ-S-763 or ASTM B582, classes 302, 303, 303se, 304 or 316, except for .040 inch (1.02 mm) flat washer.
5. Finish passivated in accordance with SAE-AMS-QQ-P-35.

FIGURE 2. Guidepin male and female.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Inches	mm
.320	8.13
.417	10.57
.812	20.62

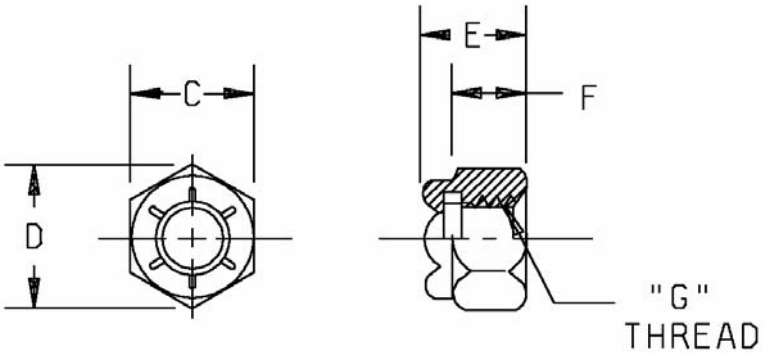
Fixed jackscrews

Fixed jackscrew designator	Gender	Connector size (number of contacts)	A Thread class 2	B Thread class 2	C dia max
M18196-1	Male	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	6-32 UNC-2A .138-32 UNC	.190 (4.83)
M18196-2	Female	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	6-32 UNC-2B .138-32 UNC	.190 (4.83)
M18196-3	Male	104	6-32 UNC-2A .138-32 UNC	8-32 UNC-2A .164-32 UNC	.137 (3.48)
M18196-4	Female	104	6-32 UNC-2A .138-32 UNC	8-32 UNC-2B .164-32 UNC	.137 (3.48)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material corrosion resistance steel in accordance with SAE-AMS-QQ-S-763 or ASTM B582, classes 302, 303, 303Se, 304 or 316, except for .040 inch (1.02 mm) flat washer.
5. Finish passivated in accordance with SAE-AMS-QQ-P-35.
6. Jackscrew may have .005 inch (0.13 mm) minimum vertical float.

FIGURE 3. Fixed jackscrews male and female.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



MINIATURE SELF-LOCKING NUT
(NYLON INSERT)

Self-locking nut dimensions

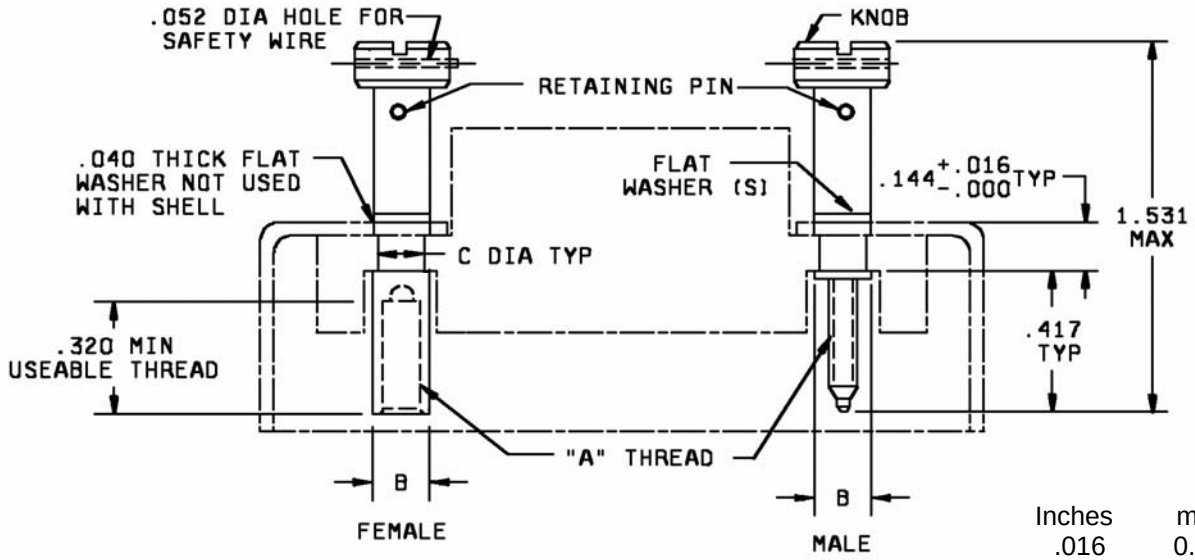
Self locking nut part number	Connector size (number of contacts)	D		E ref	F		G ref	H Thread class 3
		min	max		min	max		
MS21044-C04	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	.181 (4.06)	.190 (4.83)	.205 (5.21)	.095 (2.41)	.125 (3.18)	.070 (1.78)	4-40 UNC-3B .112-40 UNC
MS21044-C06	104	.240 (6.10)	.252 (6.04)	.275 (6.99)	.133 (3.38)	.153 (3.89)	.081 (2.06)	6-32 UNC-3B .138-32 UNC

* National Aerospace Standard NASM21044. Not to be ordered with this slash sheet (see note 1).

NOTES:
1. Self-locking nut shown for reference purposes only.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
5. Material corrosion resistance steel in accordance with SAE-AMS-QQ-S-763 or ASTM B582, classes 302, 303, 303Se, 304 or 316.
6. Finish passivated in accordance with SAE-AMS-QQ-P-35.
7. Not to be used where temperatures exceed 250°F (121°C).
8. Do not reuse nylon nut after removal.
9. Copies of NASM21044 are available online at http://www.aia-aerospace.org/, or may be purchased from Aerospace Industries Associations of America Inc., 1250 Eye Street N.W., Washington, DC 20005-3924.

FIGURE 4. Self-locking nut.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



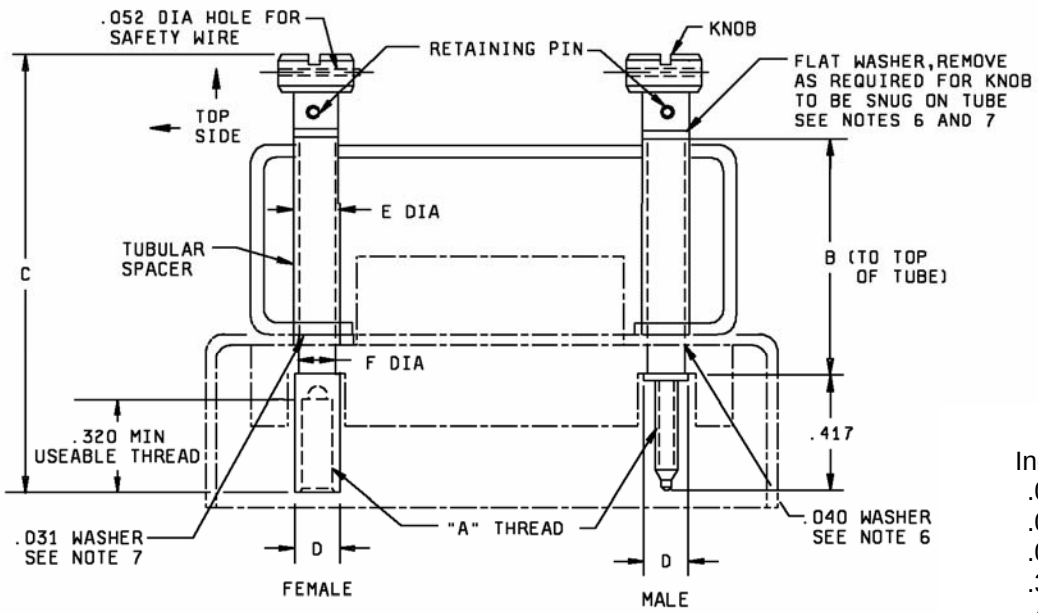
Inches	mm
.016	0.41
.040	1.02
.052	1.32
.144	3.60
.320	8.13
.417	10.59
1.531	58.89

Short turnable jackscrew designator	Gender	Connector size (number of contacts)	"A" thread class 2	"B" dia max	C dia ±.003 (0.08 mm)
M18195-1	Male	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	6-32 UNC-2A .138-32 UNC	.190 (4.83)	.115 (2.92)
M18195-2	Female	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	6-32 UNC-2B .138-32 UNC	.190 (4.83)	.115 (2.92)
M18195-3	Male	104	8-32 UNC-2A .164-32 UNC	.225 (5.72)	.137 (3.48)
M18195-4	Female	104	8-32 UNC-2B .164-32 UNC	.225 (5.72)	.137 (3.48)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material corrosion resistance steel in accordance with SAE-AMS-QQ-S-763 or ASTM B582, classes 302, 303, 303Se, 304 or 316 except for aluminum alloy knob and .040 inch (1.02 mm) flat washer.
5. Finish passivated in accordance with SAE-AMS-QQ-P-35 except for anodized knob.
6. Parts bagged disassembled.

FIGURE 5. Short turnable jackscrews

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Inches	mm
.031	0.79
.040	1.02
.052	1.32
.320	8.13
.417	10.59

MULTIPIECE CONSTRUCTION

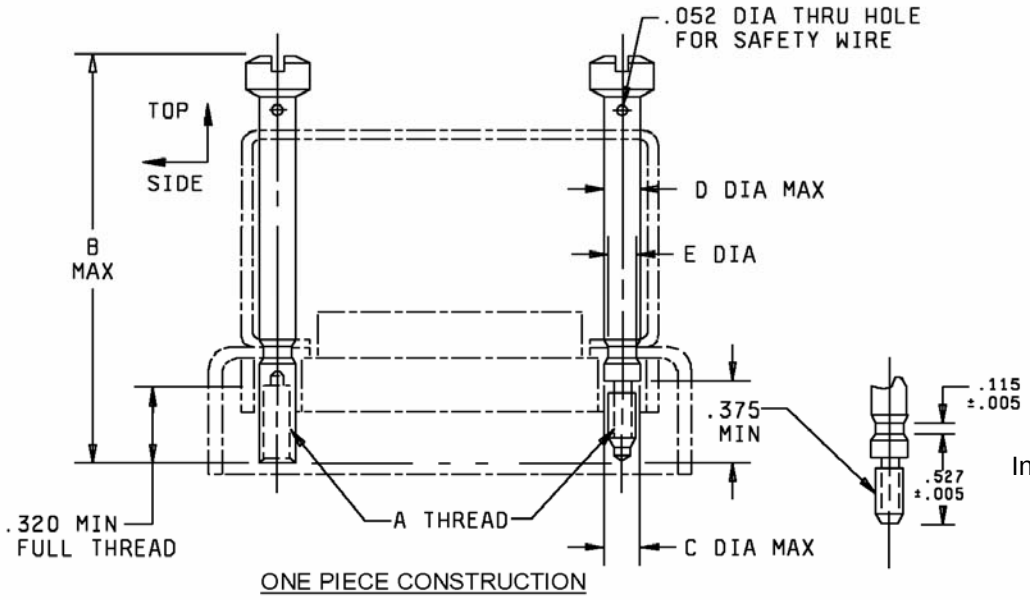
Multipiece turnable jackscrew designator	Gender	Connector size (Number of contacts)	"A" Thread class 2	"B" ±.016 (0.41 mm)	C max	D dia max	E dia ±.004 (0.10 mm)	F dia ±.003
M18194-1	Male	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	6-32 UNC-2A .138-32 UNC	1.500 (43.10)	2.467 (62.66)	.190 (4.83)	.188 (4.78)	.115 (2.92)
M18194-2	Female	7, 9, 14, 18, 20, 26, 34, 42, 50, 66, 75	6-32-UNC-2B .138-32 UNC	1.500 (43.10)	2.467 (62.66)	.190 (4.83)	.188 (4.78)	.115 (2.92)
M18194-3	Male	104	8-32-UNC-2A .164-32 UNC	2.495 (63.37)	3.438 (87.33)	.225 (5.72)	.219 (5.56)	.137 (3.48)
M18194-4	Female	104	8-32-UNC-2B .164-32 UNC	2.495 (63.37)	3.438 (87.43)	.225 (5.72)	.219 (5.56)	.137 (3.48)

For use with M18192-1 through -9, M18193-1 and -2 and M24133-1 through -4 shields and M18203-1 through -10 and M18204-1 through -10 shells.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material corrosion resistance steel in accordance with SAE-AMS-QQ-S-763 or ASTM B582, classes 302, 303, 303Se, 304 or 316 except for aluminum alloy knob and tubular, spacer.
5. Finish passivated in accordance with SAE-AMS-QQ-P-35 except for anodized knob and tubular spacer on multipiece construction.
6. Washer .040 inch thick (1.02 mm) supplied with all sizes, not used with shell.
7. Washer .031 inch thick (0.79 mm) supplied with -1 and -2 is not used with shield sizes 20 and smaller.

FIGURE 6. Multipiece turnable jackscrews

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Inches	mm
.005	0.13
.052	1.32
.115	2.92
.320	8.13
.375	9.53
.527	13.39

Long turnable jackscrew designator	Gender	Connector size (number of contacts)	"A" Thread class 2	"B" max	C dia max	D dia max	E dia ±.002 (0.05 mm)
M18194-5	Male	34, 42, 50, 66, 75	6-32-UNC-2A .138-32 UNC	2.438 (61.93)	.192 (4.88)	.192 (4.88)	.111 (2.82)
M18194-6	Female	34, 42, 50, 66, 75	6-32-UNC-2B .138-32 UNC	2.438 (61.93)	.192 (4.88)	.192 (4.88)	.111 (2.82)
M18194-7	Male	104	8-32-UNC-2A .164-32 UNC	3.282 (83.36)	.225 (5.72)	.225 (5.72)	.140 (3.56)
M18194-8	Female	104	8-32-UNC-2B .164-32 UNC	3.282 (83.36)	.225 (5.72)	.225 (5.72)	.140 (3.56)

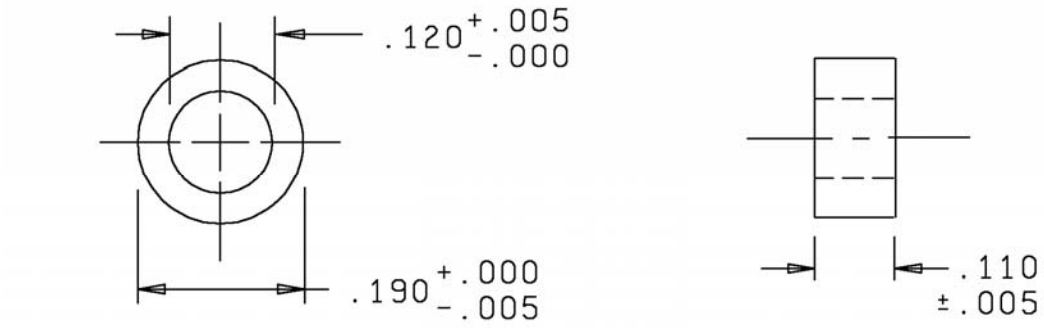
For use with M18192-10 through -13, M18193-3 and -4, and M24133-9 through -12 shields or M18203-11 through -16 and M18204-11 through -16 shells.

When M18203-11 through -16 or M18204-11 through -16 shells are required to be used with a shield, use M18192-1 through -9 or M18193-1 or -2 shield.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material: Corrosion resistant steel in accordance with SAE-AMS-QQ-S-763 or ASTM B582, classes 302, 303, 303Se, 304, or 316 except for aluminum alloy knob.
5. Finish passivated in accordance with SAE-AMS-QQ-P-35 except for anodized knob.

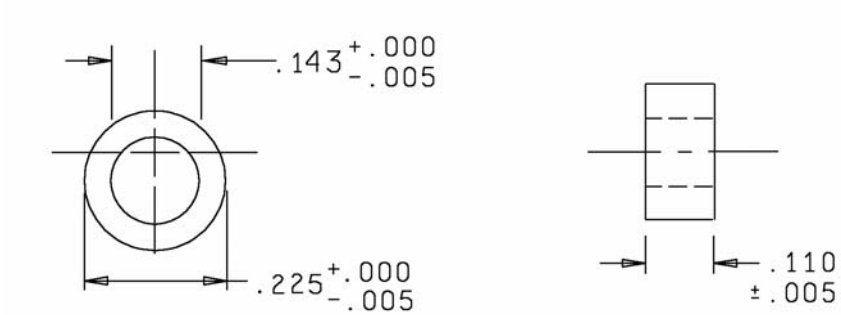
FIGURE 7. Long turnable jackscrew one piece construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



For use with inserts sizes 34, 42, 50, 66, and 75

Designator M18173-1



For use with insert size 104

Designator M18173-2

Inches	mm
.005	0.13
.110	2.79
.120	3.05
.143	3.63
.190	4.83
.225	5.71

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances is ±.015 inch (0.38 mm).
4. Washer: Corrosion resistant steel, in accordance with SAE-AMS-QQ-S-763 or ASTM B582, classes 302, 303, 303Se, 304, or 316.
5. Washer finish: Passivated in accordance with SAE-AMS-QQ-P-35 and self-locking nut.

FIGURE 8. Flat washer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

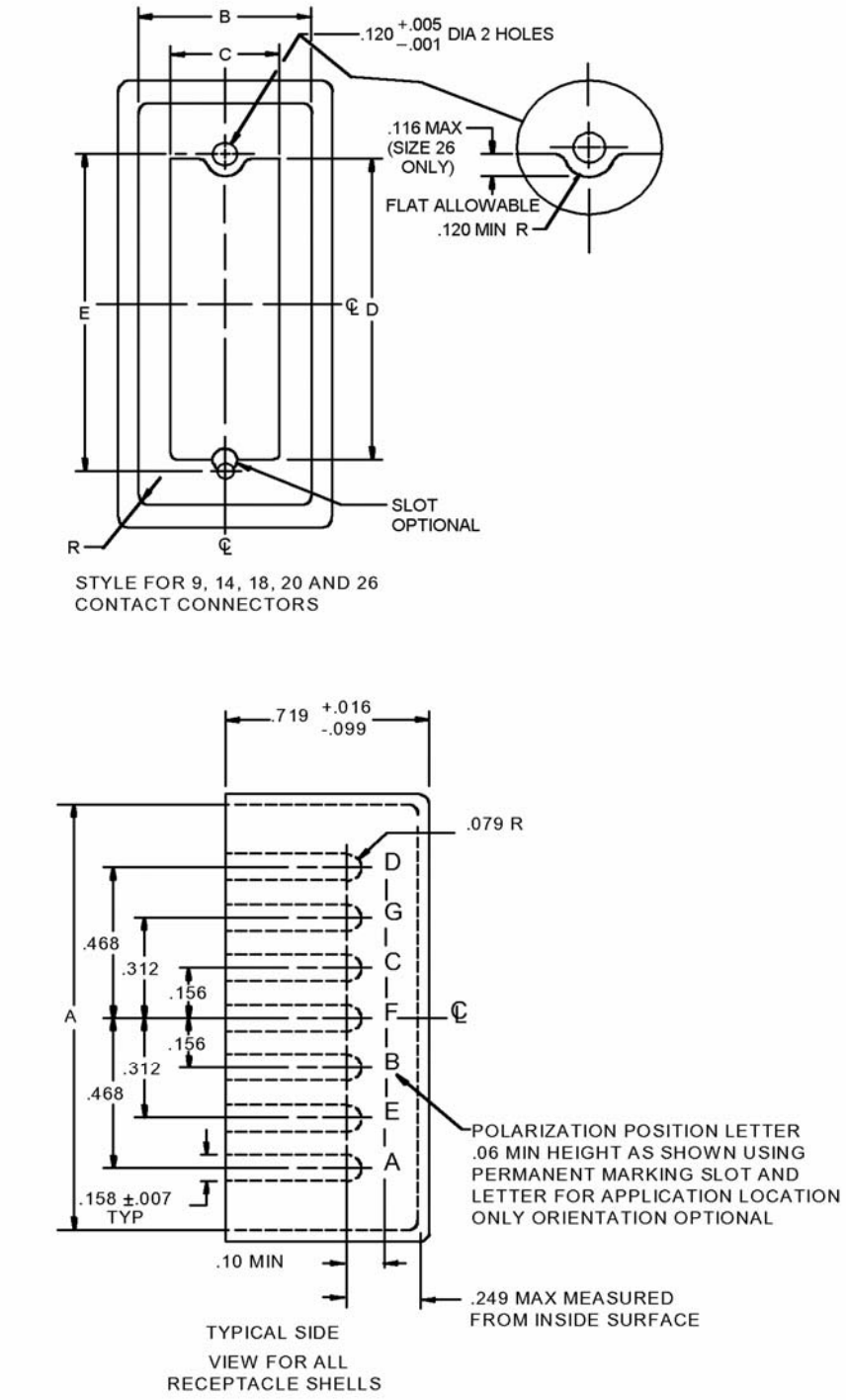
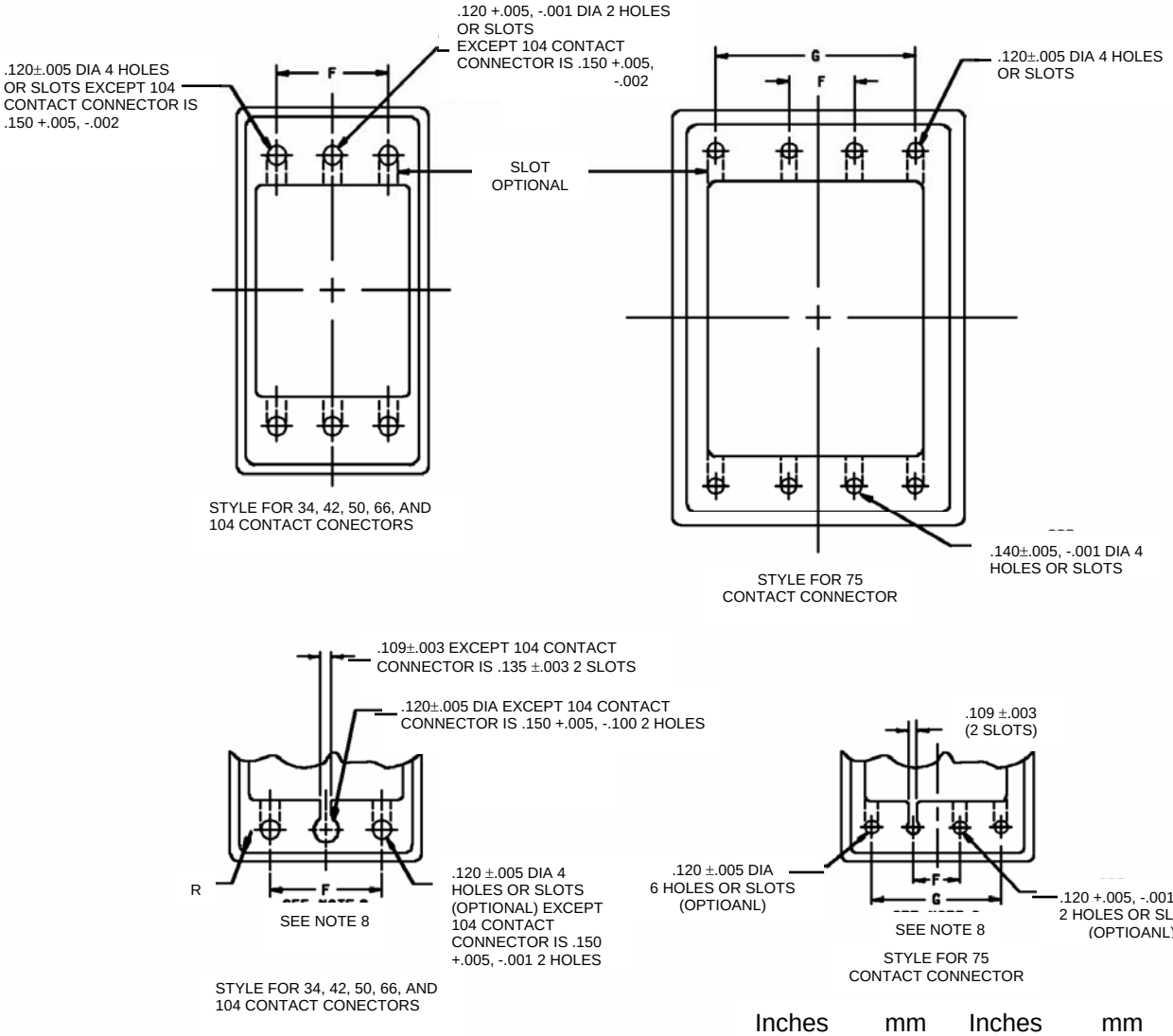


FIGURE 9. Receptacle shell, polarizing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm
.001	0.03	.116	2.95
.002	0.05	.120	3.05
.003	0.08	.135	3.43
.005	0.13	.140	3.56
.007	0.18	.150	3.81
.016	0.41	.156	3.96
.06	1.5	.158	4.01
.079	2.01	.249	6.32
.099	2.51	.312	7.92
.10	2.5	.468	11.89
.109	2.77	.719	18.26

Receptacle shell M18204, polarizing

FIGURE 9. Receptacle shell, polarizing - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

Receptacle shell with polarization designator	Connector size (number of contacts)	A min	B min	C min	D min	E ±.005 (0.13 mm)	F ±.005 (0.13 mm)	G ±.005 (0.13 mm)	R
M18204-9( )	9	1.448 (36.78)	.512 (13.00)	.375 (9.52)	.864 (21.94)	1.000 (25.40)	---	---	.141 (3.58)
M18204-1( )	14	1.388 (35.36)	.575 (14.60)	.440 (11.18)	.800 (20.32)	.938 (23.82)	---	---	.141 (3.58)
M18204-17( )	18	1.450 (36.80)	.696 (17.68)	.565 (14.35)	.880 (22.40)	1.00 (25.40)	---	---	.141 (3.58)
M18204-2( )	20	1.702 (43.23)	.575 (14.60)	.440 (11.18)	1.097 (27.86)	1.250 (31.75)	---	---	.141 (3.58)
M18204-10( )	26	1.763 (44.78)	.696 (17.68)	.565 (14.35)	1.182 (30.02)	1.312 (33.32)	---	---	.141 (3.58)
M18204-3( )	34	2.138 (54.30)	.888 (22.56)	.745 (18.92)	1.400 (35.76)	1.688 (42.88)	.468 (11.89)	---	.141 (3.58)
M18204-4( )	42	2.450 (62.23)	.888 (22.56)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)	---	.141 (3.58)
M18204-5( )	50	2.856 (72.54)	1.015 (25.78)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)	---	.188 (4.78) (see note 4)
M18204-6( )	66	2.420 (61.47)	1.264 (32.11)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01	.500 (12.70)	---	.375 (9.52)
M18204-7( )	75	2.855 (72.52)	1.370 (34.80)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.203 (5.16)
M18204-8( )	104	2.900 (73.66)	1.680 (42.67)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)	---	.422 (10.72)
M18204-11( ) see note 9	34	2.138 (54.30)	.888 (22.56)	.745 (18.92)	1.400 (35.56)	1.688 (42.88)	.468 (11.89)	---	.141 (3.58)
M18204-12( ) see note 9	42	2.450 (62.23)	.888 (22.56)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)	---	.141 (3.58)
M18204-13( ) see note 9	50	2.856 (72.54)	1.015 (25.78)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)	---	.188 (4.78) (see note 4)
M18204-14( ) see note 9	66	2.420 (61.47)	1.264 (32.10)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01)	.500 (12.70)	---	.375 (9.52)
M18204-15( ) see note 9	75	2.855 (72.52)	1.370 (34.80)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.203 (5.16)
M18204-16( ) see note 9	104	2.900 (73.66)	1.680 (42.67)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)	---	.422 (10.72)

Polarization designators ( ) indicate location of polarization slot designator A through G (see note 8).

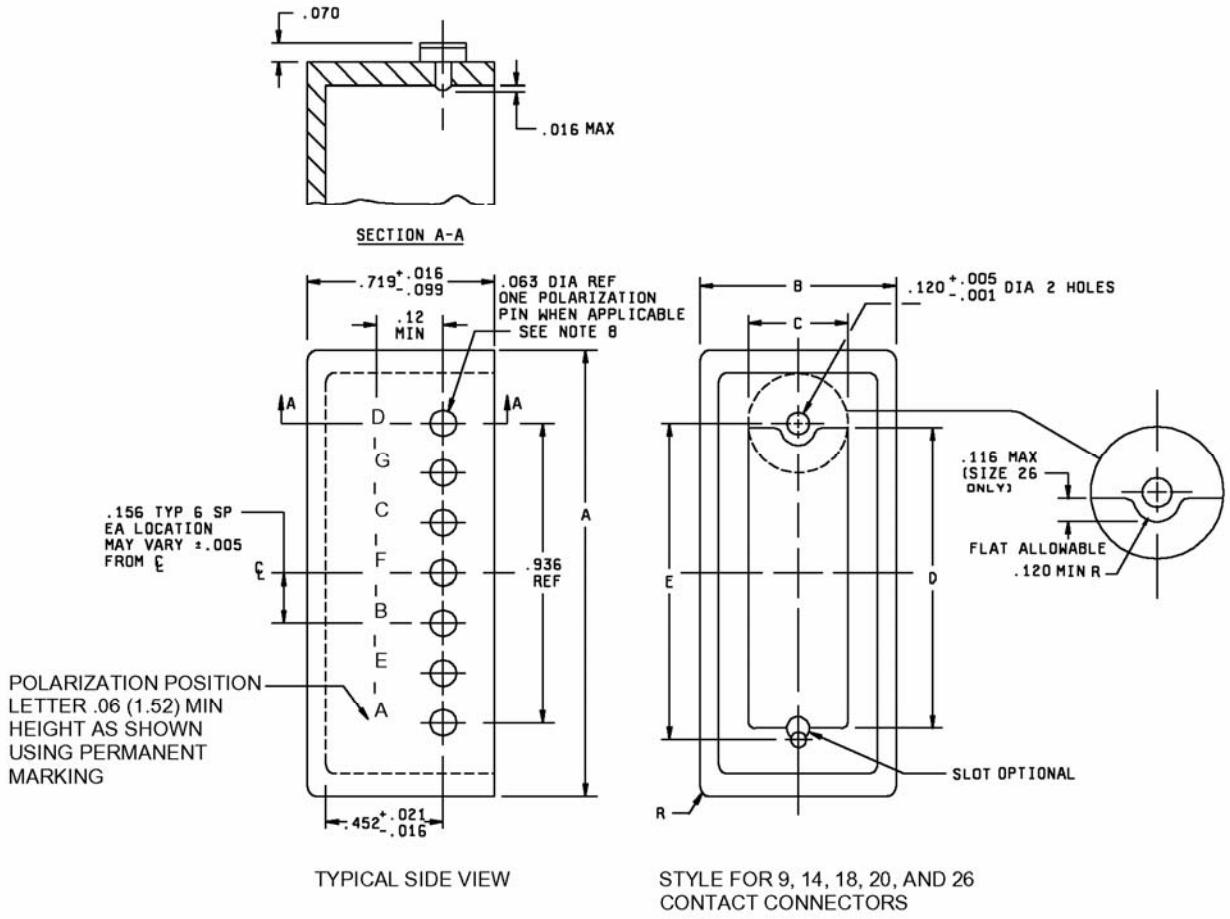
FIGURE 9. Receptacle shell polarizing - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Tolerance +.020 inch (0.51 mm) -.016 inch (0.41 mm).
5. Material aluminum alloy in accordance with specification SAE-AMS-QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
6. Finish anodize in accordance with MIL-A-8625, type I or type II.
7. Each slot may vary ±.005 inch (0.13 mm) from centerline.
8. PIN examples:
   a. Polarization required, specify as M28748/17M18204-# (A through G). Example: M28748/17M18204-5D.
   b. Polarization not required, specify as M28748/17M18204-#. Example: M28748/17M18204-5.
9. Receptacle shell designators M18204-11, -12, -13, -14, -15, and -16 are not used with one-piece jackscrew shields of M24133.

FIGURE 9. Receptacle shell polarizing - Continued.

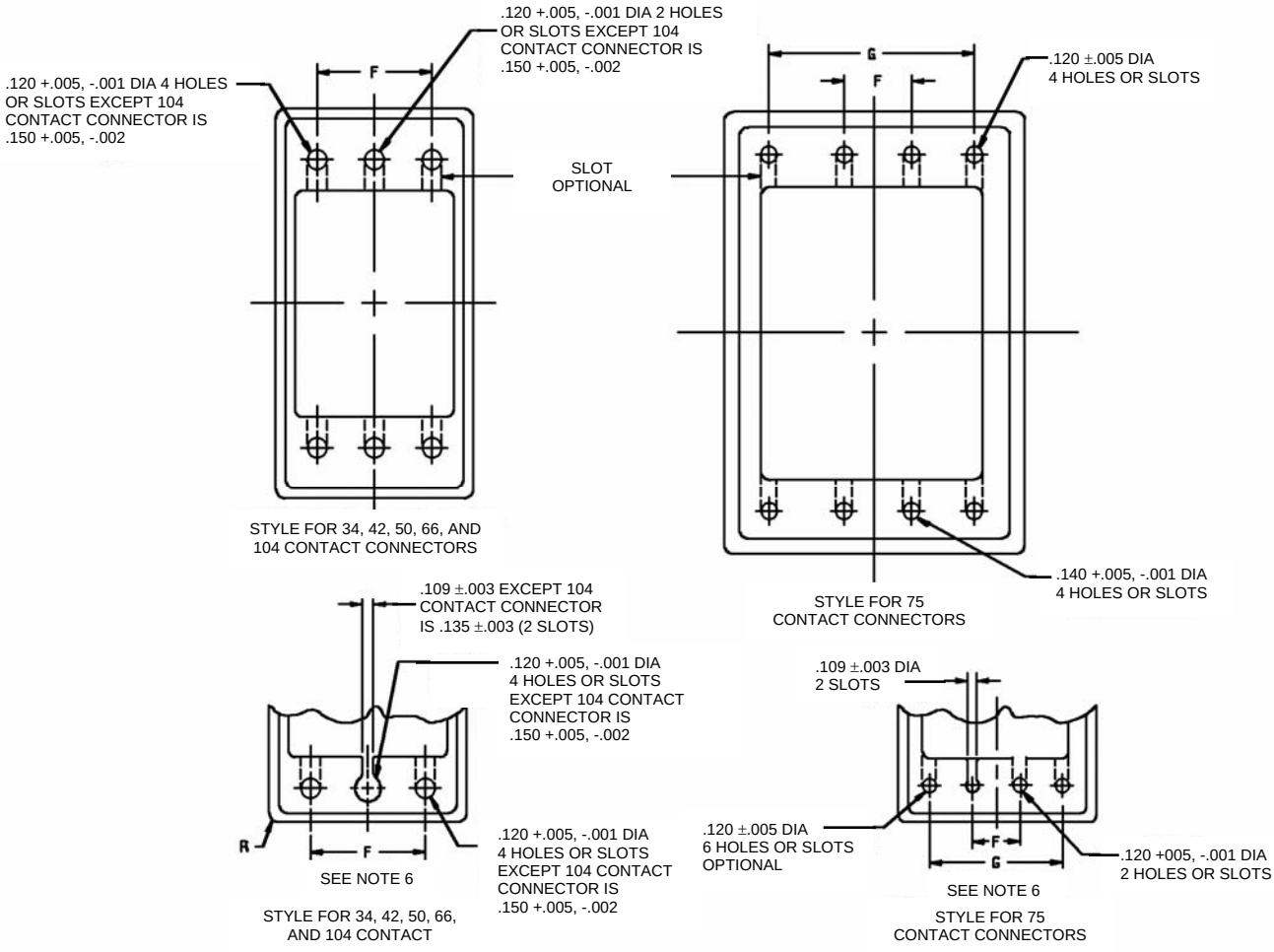
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Plug shell polarizing

FIGURE 10. Plug shell, configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm
.001	0.03	.116	2.95
.002	0.05	.12	3.0
.003	0.08	.120	3.05
.005	0.13	.135	3.43
.016	0.41	.140	3.56
.021	0.53	.150	3.81
.063	1.60	.156	3.96
.070	1.78	.234	5.94
.099	2.51	.452	11.48
.109	2.77	.719	18.26
		.936	23.77

Plug shell M18203, polarizing

FIGURE 10. Plug shell configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

Designator see note 6	Connector size (number of contacts)	A max	B max	C min	D min	E ±.005 (0.13 mm)	F ±.005 (0.13 mm)	G ±.005 (0.13 mm)	R
M18203-9( )	9	1.440 (36.58)	.500 (12.70)	.375 (9.52)	.864 (21.94)	1.000 (25.40)	---		.141 (3.58)
M18203-1( )	14	1.380 (35.05)	.567 (14.40)	.440 (11.18)	.800 (20.32)	.938 (23.82)	---		.141 (.538)
M18203-17( )	18	1.440 (36.58)	.680 (14.70)	.580 (14.70)	.880 (22.40)	1.000 (25.40)	---	---	.141 (.538)
M18203-2( )	20	1.700 (43.18)	.567 (14.40)	.440 (11.18)	1.097 (27.86)	1.250 (31.75)	---		.141 (.538)
M18203-10( )	26	1.755 (44.58)	.688 (17.48)	.565 (14.35)	1.182 (30.02)	1.312 (33.32)	---		.141 (3.58)
M18203-3( )	34	2.130 (54.10)	.880 (22.35)	.745 (18.92)	1.400 (35.76)	1.688 (42.88)	.468 (11.89)		.141 (3.58)
M18203-4( )	42	2.442 (62.03)	.880 (22.35)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)		.141 (3.58)
M18203-5( )	50	2.846 (72.39)	1.005 (25.53)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)		.188 (4.78)
M18203-6( )	66	2.410 (61.21)	1.254 (31.85)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01	.500 (12.70)		.375 (9.52)
M18203-7( )	75	2.850 (72.39)	1.365 (34.67)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.187 (4.75)
M18203-8( )	104	2.890 (73.41)	1.670 (42.42)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)		.422 (10.72)
M18203-11( ) see note 8	34	2.130 (54.10)	.880 (22.35)	.745 (18.92)	1.400 (35.56)	1.688 (42.88)	.468 (11.89)		.141 (3.58)
M18203-12( ) see note 8	42	2.442 (62.03)	.880 (22.35)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)		.141 (3.58)
M18203-13( ) see note 8	50	2.846 (72.29)	1.005 (25.53)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)		.188 (4.78)
M18203-14( ) see note 8	66	2.410 (61.21)	1.254 (31.85)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01)	.500 (12.70)		.375 (9.52)
M18203-15( ) see note 8	75	2.850 (72.39)	1.365 (34.67)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.187 (4.75)
M18203-16( ) see note 8	104	2.890 (73.41)	1.670 (42.42)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)		.422 (10.72)

Polarization designators ( ) indicate location of polarization plug A through G (see note 8).

Plug shell M18203, polarizing

FIGURE 10. Plug shell configurations - Continued.

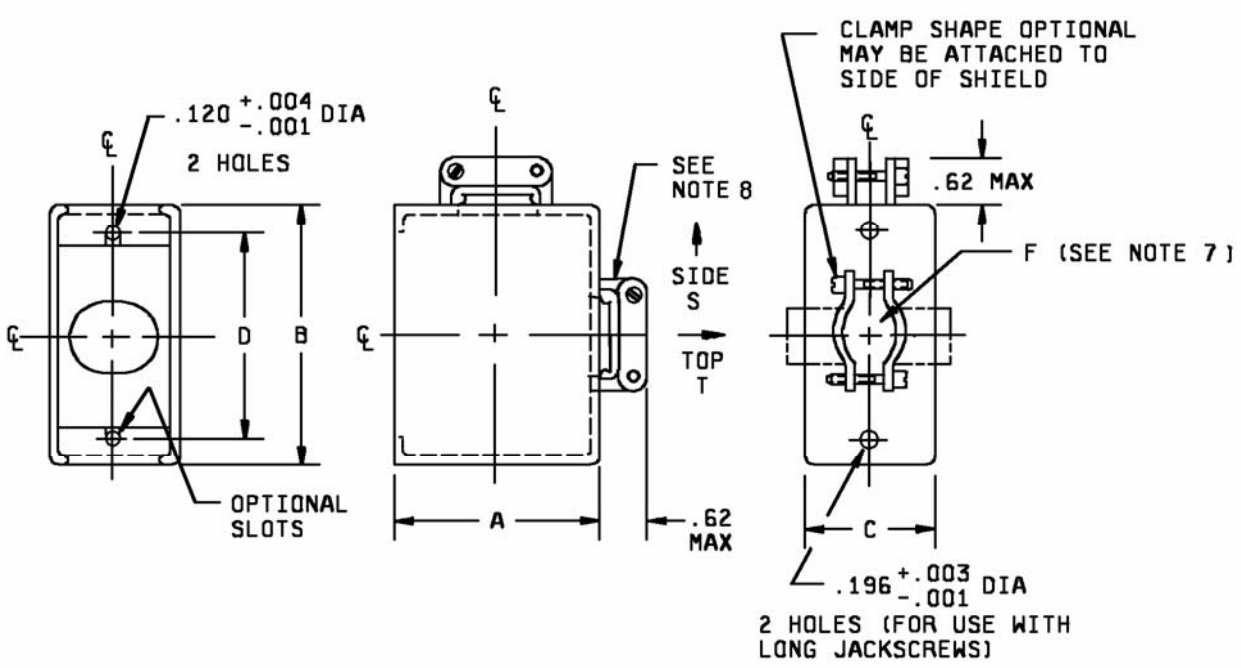
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with SAE-AMS-QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.
6. PIN examples:
   a. Polarization required, specify as M28748/17M18203-#(A through G): Example: M28748/17M18203-14D (see note 7).
   b. Polarization not required, specify as M28748/17M18203-#: Example: M28748/17M18203-5.
7 One polarization pin is used when applicable. Polarization pin will be packaged lose. If a polarization pin only is required order separately as M28748/17M18205.
8. M28748/17M18203-11, -12, -13, -14, -15, and -16 are not used with one-piece jackscrew shields of M28748/17M24133.

Plug shell polarizing

FIGURE 10. Plug shell configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



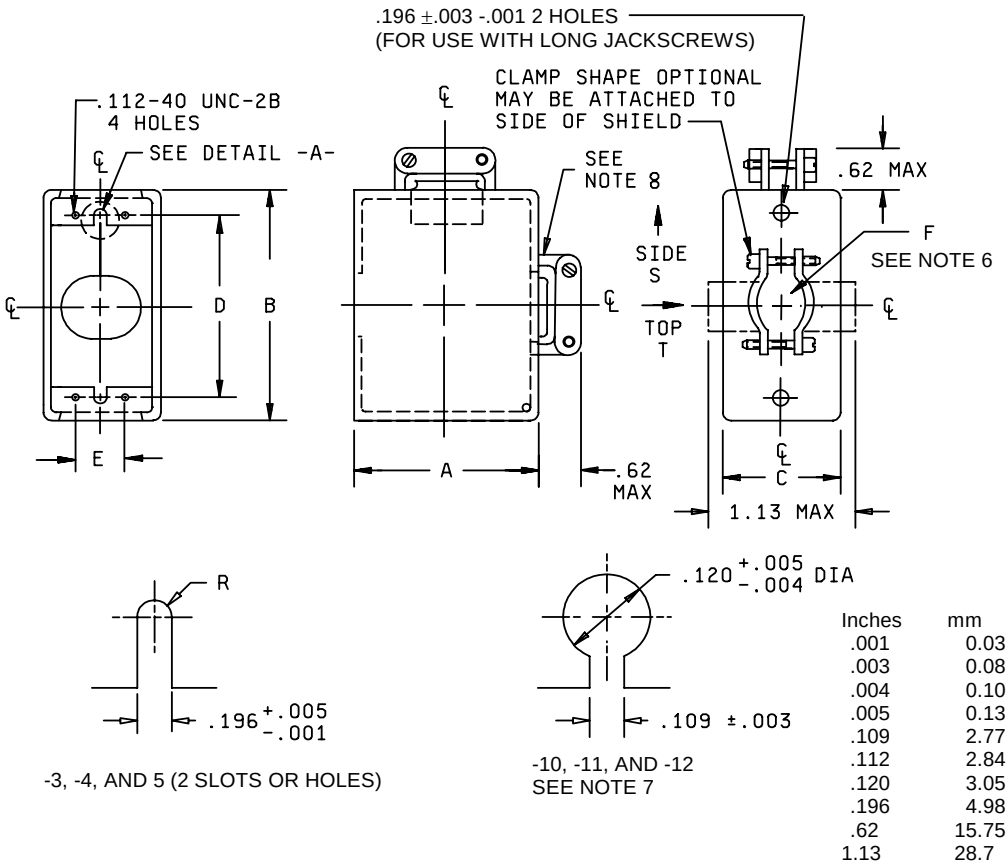
Inches	mm
.001	0.03
.003	0.08
.004	0.10
.120	3.05
.196	4.98
.62	15.75

Shield designator 1/	Shield and shield clamp location	Connector size (number of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	F Minimum cable opening square inch
M18192-T7 M18192-S7	top opening side opening	7	1.18 (29.97) 1.35 (34.29)	1.220 (30.99)	.437 (11.10)	.906 (23.01)	.18 (4.57)
M18192-T8 M18192-S8	top opening side opening	9	1.18 (29.97) 1.35 (34.29)	1.312 (33.32)	.437 (11.10)	1.000 (25.40)	.10 (2.54)
M18192-T1 M18192-S1	top opening side opening	14	1.18 (29.97) 1.35 (34.29)	1.250 (31.75)	.500 (12.70)	.938 (23.83)	.19 (4.83)
M18192-T14 M18192-S14	top opening side opening	18	1.18 (29.97) 1.35 (34.29)	1.190 (30.20)	.630 (16.00)	1.000 (25.40)	.22 (5.59)
M18192-T2 M18192-S2	top opening side opening	20	1.18 (29.97) 1.35 (34.29)	1.562 (39.67)	.500 (12.70)	1.250 (31.75)	.22 (5.59)
M18192-T9 M18192-S9	top opening side opening	26	1.18 (29.97) 1.35 (34.29)	1.625 (41.28)	.640 (16.26)	1.312 (33.32)	.22 (5.59)

1/ When used with MIL-DTL-28748/5 and 6 slash sheets, the shield is only used when shell M18203 or M18204 is specified.

FIGURE 11. Shield 7, 9, 14, 20, 26, 34, 42, 50, and 75 contact positions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

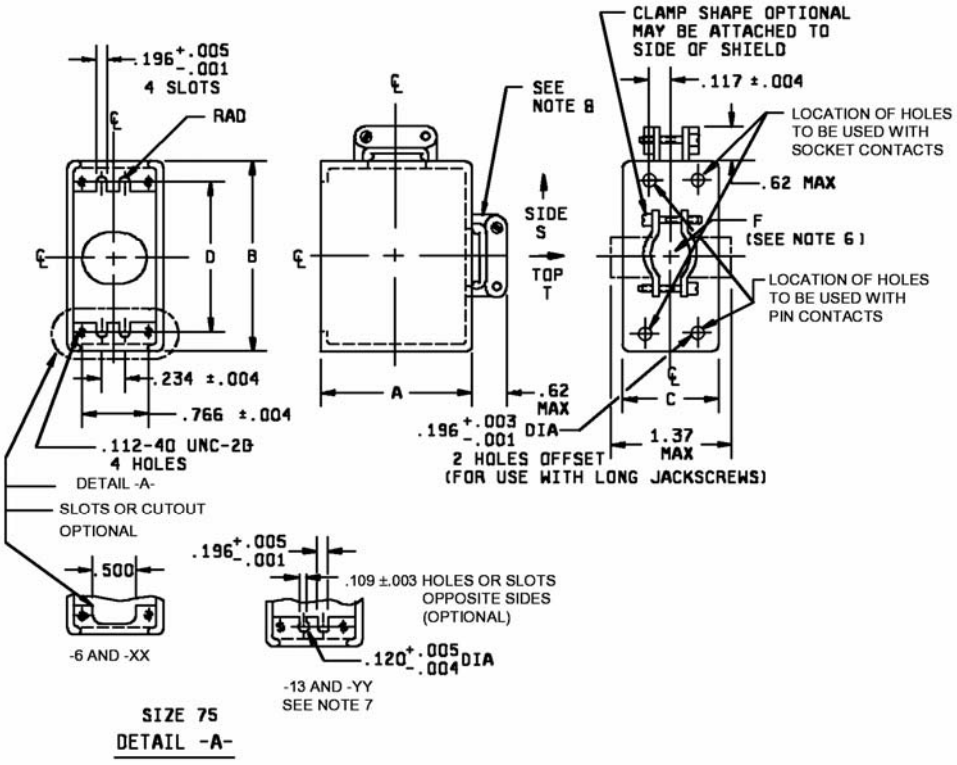


DETAIL -A-

Shield designator	Shield and shield clamp location suffix designator	Connector size (number of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	E ±.010 (0.25)	F Minimum cable opening square inch
M18192-T3 M18192-S3	top opening side opening	34	1.22 (30.99) 1.35 (34.29)	2.000 (50.80)	.828 (21.03)	1.688 (42.88)	.468 (11.89)	.34 (8.64)
M18192-T4 M18192-S4	top opening side opening	42	1.22 (30.99) 1.35 (34.29)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.468 (11.89)	.34 (8.64)
M18192-T5 M18192-T5	top opening side opening	50	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	.828 (21.03)	2.282 (57.96)	.468 (11.89)	.34 (8.64)
M18192-T10 M18192-S10	top opening side opening	34	1.22 (30.99) 1.35 (34.29)	2.000 (50.80)	.828 (21.03)	1.688 (42.88)	.468 (11.89)	.34 (8.64)
M18192-T11 M18192-S11	top opening side opening	42	1.22 (30.99) 1.35 (34.29)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.468 (11.89)	.34 (8.64)
M18192-T12 M18192-S12	top opening side opening	50	1.22 (30.99) 1.55 (39.37)	2.594 (65.89)	.828 (21.03)	2.282 (57.96)	.468 (11.89)	.34 (8.64)

FIGURE 11. Shields for 7, 9, 14, 20, 26, 34, 42, 50, and 75 contact positions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



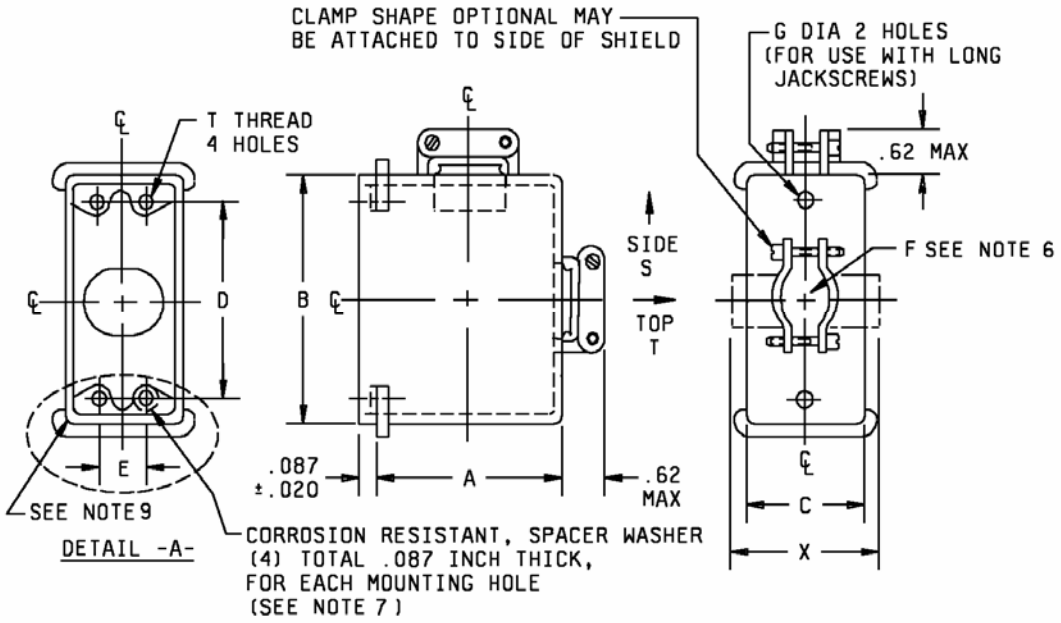
Inches	mm
.001	0.03
.003	0.08
.004	0.10
.005	0.13
.109	2.77
.112	2.84
.117	2.97
.120	3.05
.196	4.98
.234	5.94
.500	12.70
.62	15.75
.766	19.46
1.37	34.8

Designator	Shield and shield clamp location suffix designator	Connector size (number of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	F Minimum cable opening square inch
M18192-T6 M18192-S6	top opening side opening	75 Pin only	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (65.89)	2.282 (57.96)	.55 (13.97)
M18192-T13 M18192-S13	top opening side opening	75 Pin only	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (65.89)	2.282 (57.96)	.55 (13.97)
M18192-TXX M18192-SXX	top opening side opening	75 Socket only	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (65.89)	2.282 (57.96)	.55 (13.97)
M18192-TYY M18192-SYY	top opening side opening	75 Socket only	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (65.89)	2.282 (57.96)	.55 (13.97)

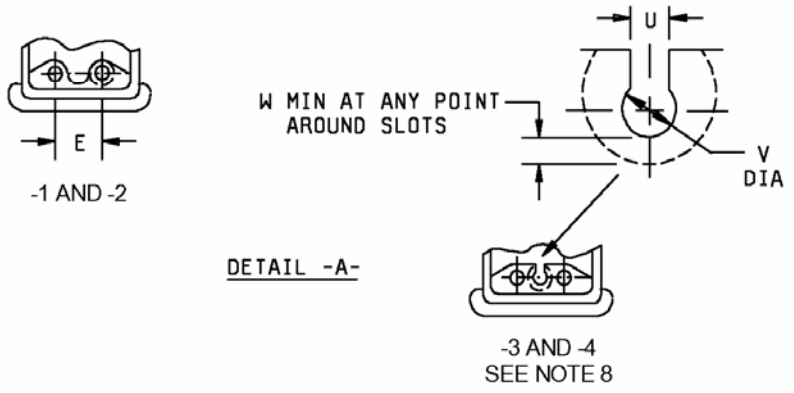
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with SAE-AMS-QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodized in accordance with MIL-A-8625, type I or type II.
6. Edges of cable entry hole to be rounded to prevent insulation chafing.
7. M18192-T10, S10, T11, S12, T13, S13, TYY, and SYY are for use with one-piece jackscrews.
8. Clamp hinge may attach to side of shield.

FIGURE 11. Shields for 7, 9, 14, 20, 26, 34, 42, 50, and 75 contact positions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



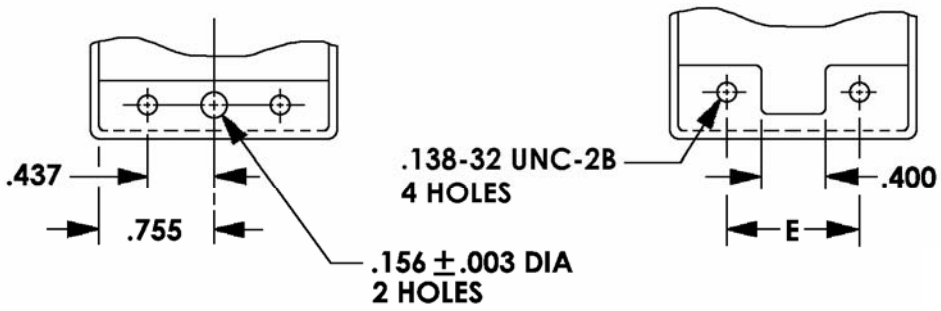
Shield designator	Shield and shield clamp location (see note 8)	U ±.003 (0.08 mm)	V +.005 (0.13 mm) -.001 (0.03 mm)	W Min	Connector size (number of contacts)
M18193-T3 M18193-S3	top opening side opening	.109 (2.77)	.120 (3.05)	.062 (1.57)	66
M18193-T4 M18193-S4	top opening side opening	.133 (3.38)	.150 (3.81)	.110 (2.79)	104



Inches	mm
.020	0.51
.087	2.21
.62	15.75

FIGURE 12. Shields for 66 and 104 contact connector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



ALTERNATE CONFIGURATIONS –2 AND -4

Inches	mm
.003	0.08
.138	3.51
.156	3.96
.400	10.16
.437	11.10
.755	19.18

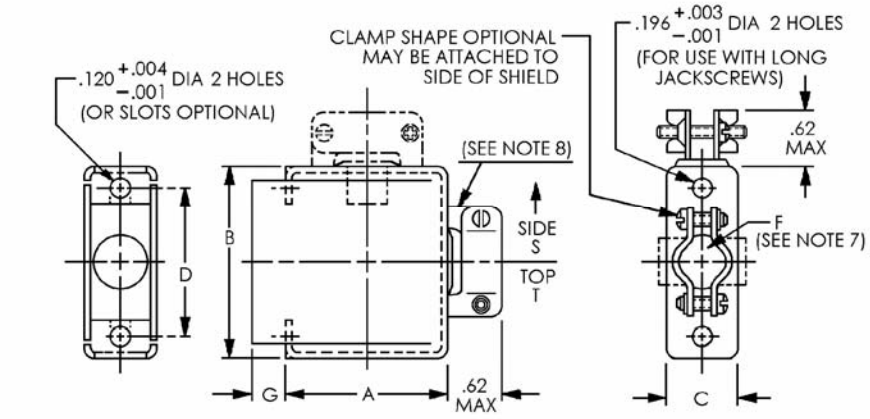
Shield designator	Shield and shield clamp location	Connector size (number of contacts)	A ±.03 (0.76 mm)	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	E ±.005 (0.13 mm)	F Minimum cable opening sq. inch	G +.003 (0.08 mm) -.001 (0.03 mm)	T thread	X max
M18193-T1 M18193-S1	top opening side opening	66	1.17 (29.72)	2.366 (60.10)	1.211 (30.76)	1.969 (50.01)	.500 (12.70)	.55 (13.97)	.196 (4.98)	.112-40 UNC-2B	1.37 (34.80)
M18193-T2 M18193-S2	top opening side opening	104	2.23 (56.64)	2.835 (72.01)	1.625 (41.28)	2.375 (60.33)	.875 (22.23)	1.10 (27.94)	.229 (5.82)	.138-32 UNC-2B	1.88 (47.75)
M18193-T3 M18193-S3	top opening side opening	66	1.17 (29.72)	2.366 (60.10)	1.211 (30.76)	1.969 (50.01)	.500 (12.70)	.55 (13.97)	.196 (4.98)	.112-40 UNC-2B	1.37 (34.80)
M18193-T4 M18193-S4	top opening side opening	104	2.23 (56.64)	2.835 (72.01)	1.625 (41.28)	2.375 (60.33)	.875 (22.23)	1.10 (27.94)	.229 (5.82)	.138-32 UNC-2B	1.88 (47.75)

M18193-T3, S3, T4, and S4 are used with one-piece jackscrews.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with SAE-AMS-QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodized in accordance with MIL-A-8625, type I or type II.
6. Edges of cable entry hole to be rounded to prevent insulation chafing.
7. Spacer washers (4) used only with shell. Discard otherwise. Aluminum shall be in accordance with SAE-AMS-QQ-A250/11, T6, anodized in accordance with MIL-A-8625, type I or II.
8. M18193-T3, S3, T4, S4 are for use with one-piece jackscrews.
9. No crack, chips, or sharp edges on all 4 edges of flange.

FIGURE 12. Shields for 66 and 104 contact connector - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



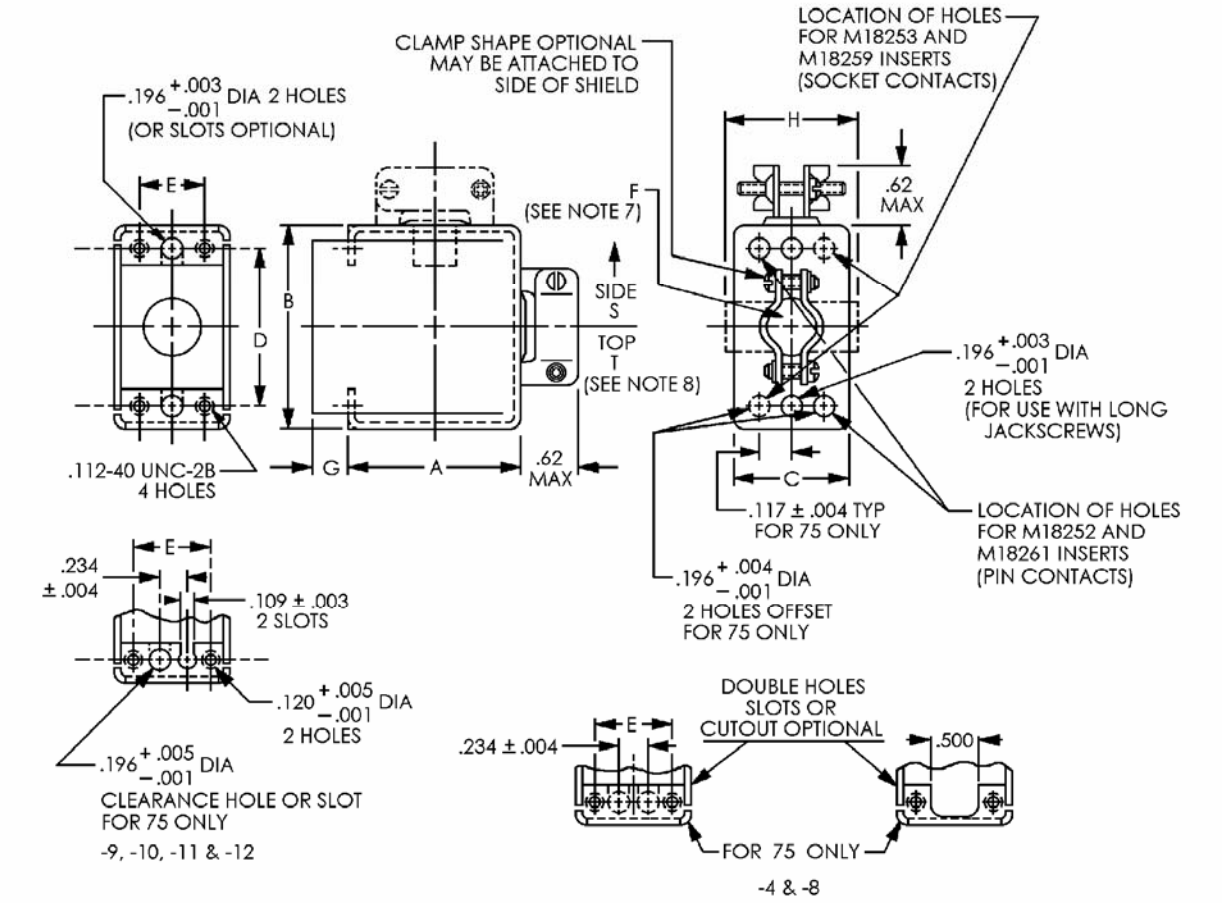
Shield designator	Connector size (number of contacts)	A min max	B ±0.016 (0.41) (mm)	C ±0.016 (0.41) (mm)	D ±0.005 (0.13) (mm)	F Minimum cable opening square inch (mm)	G ±0.047 (1.19) (mm)
M24132-1T M24132-1S	7	1.18 (29.97) 1.35 (34.3)	1.218 (32.51)	0.437 (11.10)	0.906 (23.01)	0.07 (1.8)	0.265 (6.73)
M24132-2T M24132-2S	9	1.18 (29.97) 1.35 (34.3	1.312 (33.32)	0.437 (11.10)	1.000 (25.40)	0.16 (4.1)	0.265 (6.73)
M24132-3T M24132-3S	14	1.18 (29.97) 1.35 (34.3	1.250 (31.75)	0.500 (12.70)	0.937 (23.80)	0.19 (4.82)	0.265 (6.73)
M24132-4T M24132-4S	20	1.18 (30.0) 1.35 (34.3)	1.562 (39.67)	0.500 (12.70)	1.250 (31.75)	0.22 (5.6)	0.265 (6.73)
M24132-5T M24132-5S	26	1.18 (30.0) 1.35 (34.3)	1.625 (41.28)	0.634 (16.10)	1.312 (33.32)	0.26 (6.6)	0.265 (6.73)

The following designators are the same as those listed above, except the two .120 inch (3.05 mm) diameter thru holes for mounting are threaded and the two .196 inch (4.98 mm) diameter holes are omitted. NOT RECOMMENDED FOR NEW DESIGN.

Shield designator	Connector size (number of contacts)	Thread size
M24132-1T M24132-1S	7	4-40 UNC-2B .112-40 UNC-2B
M24132-2T M24132-2S	9	4-40 UNC-2B .112-40 UNC-2B
M24132-3T M24132-3S	14	4-40 UNC-2B .112-40 UNC-2B
M24132-4T M24132-4S	20	4-40 UNC-2B .112-40 UNC-2B
M24132-5T M24132-5S	26	4-40 UNC-2B .112-40 UNC-2B

FIGURE 13. Shields for 7, 9, 14, 20, 26, 34, 42, 50, and 75.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Shield designator	Connector size (number of contacts)	A min max	B ±0.016 (0.41) (mm)	C ±0.016 (0.41) (mm)	D ±0.005 (0.13) (mm)	E ±.005 (0.13) (mm)	F Minimum cable opening square inch (mm)	G ±0.047 (1.19) (mm)
M24133-T1 M24133-S1	34	1.22 (30.99) 1.35 (34.29)	2.000 (50.80)	0.828 (21.03)	1.688 (42.87)	4.68 (11.88)	.34 (8.6)	0.265 (6.73)
M24133-T2 M24133-S2	42	1.18 (29.97) 1.35 (34.29)	2.312 (58.72)	0.828 (21.03)	2.000 (50.80)	4.68 (11.88)	.34 (8.6)	0.093 (2.36)
M24133-T3 M24133-S3	50	1.22 (30.99) 1.35 (34.29)	2.594 (65.88)	0.828 (21.03)	2.282 (57.96)	4.68 (11.88)	.34 (8.6)	0.093 (2.36)
M24133-T4 M24133-S4	75	1.22 (31.0) 1.35 (34.29)	2.594 (65.88)	1.188 (30.17)	2.282 (57.96)	.766 (19.46)	.55 14.0)	0.093 (2.36)

The following designators are the same as those listed above, except the two .196 inch (4.98 mm) diameter thru holes for mounting are threaded and the two .196 inch (4.98 mm) diameter holes are omitted. NOT RECOMMENDED FOR NEW DESIGN.

FIGURE 13. Shields for 7, 9, 14, 20, 26, 34, 42, 50, and 75 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

Shield designator	Connector size (number of contacts)	Thread size
M24133-T5 M24133-S5	34	4-40 UNC-2B .112-40 UNC-2B
M24133-T6 M24133-S6	42	4-40 UNC-2B .112-40 UNC-2B
M24133-T7 M24133-S7	50	4-40 UNC-2B .112-40 UNC-2B
M24133-T8 M24133-S8	75	4-40 UNC-2B .112-40 UNC-2B

The following shields are for use with one piece jackscrews are and not used with shells.

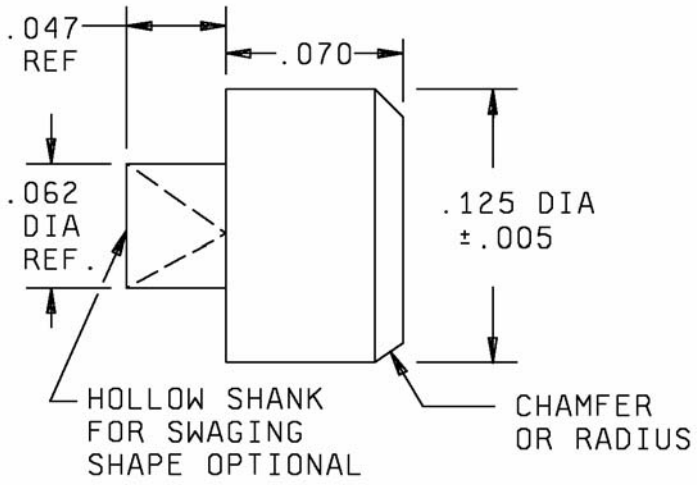
Shield designator	Connector size (number of contacts)	A min max	B ±0.016 (0.41) (mm)	C ±0.016 (0.41) (mm)	D ±0.005 (0.13) (mm)	E ±.005 (0.13) (mm)	F Minimum cable opening square inch (mm)	G ±0.047 (1.19) (mm)
M24133-T9 M24133-S9	34	1.22 (31.0) 1.35 (34.3)	2.000 (50.80)	0.828 (21.03)	1.688 (42.87)	.468 (11.89)	.34 (8.6)	.265 (6.73)
M24133-T10 M24133-S10	42	1.22 (31.0) 1.35 (34.3)	2.312 (58.72)	0.828 (21.03)	2.000 (50.80)	.468 (11.89)	.34 (8.6)	.093 (2.36)
M24133-T11 M24133-S11	50	1.22 (31.0) 1.35 (34.3)	2.594 (65.88)	0.828 (21.03)	2.282 (57.96)	.468 (11.89)	.34 (8.6)	.093 (2.36)
M24133-T12 M24133-S4	75	1.22 (31.0) 1.35 (34.3)	2.594 (65.88)	1.188 (30.17)	2.282 (57.96)	.766 (19.46)	.55 14.0)	.093 (2.36)

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.109	2.77	.120	3.05
.003	0.08	.112	2.84	.196	4.98
.004	0.10	.1120	2.845	.234	5.94
.005	0.13	.117	2.97	.500	12.70
				.62	15.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with SAE-AMS-QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.
6. Spacer washers (4) used only with shell: Discard otherwise.
7. Edges of cable entry hole to be rounded to prevent insulation chafing.
8. No cracks, chips, or sharp edges on all 4 edges of flange.

FIGURE 13. Shields for 7, 9, 14, 20, 26, 34, 42, 50, and 75 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Pin polarizing, straight headed.

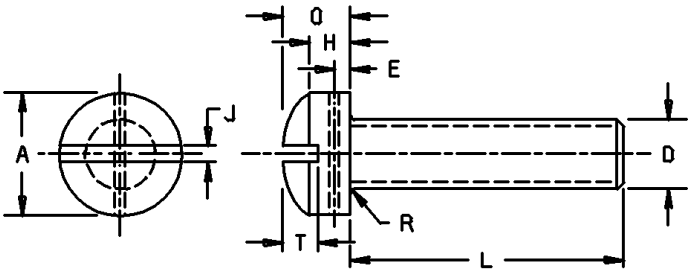
Inches	mm
.005	0.13
.047	1.19
.062	1.57
.070	1.78
.125	3.18

Designator M18205

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.016 inch (0.41 mm).
3. Metric equivalents are given for information only.
4. Material: Corrosion resistant steel in accordance with SAE-AMS-QQ-S-763, classes 302, 303, 303Se, 304, 316, or ASTM A532.
5. Passivated in accordance with SAE-AMS-QQ-P-35.

FIGURE 14. Pin polarizing configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



These are machined drilled fillister head screws. Screws furnished with shields when required, crossed drilled hole is optional.

NASM35275 PIN 1/	Nominal size	Threads per inch, UNC-2A (Pitch)	D Body diameter min	D Body diameter max	A Head diameter min	A Head diameter max	H Head side height min	H Head side height max	O Total head height min	O Total head height max
MS35275-215	.112 (#4) (2.845)	40 (.64)	.0925 (2.350)	.1120 (2.845)	.166 (4.216)	.183 (4.648)	.072 (1.829)	.079 (2.007)	.094 (2.388)	.107 (2.718)
MS35255-229	.138 (#6) (3.505)	32 (.79)	.1141 (2.898)	.1380 (3.505)	.208 (5.283)	.226 (5.740)	.089 (2.261)	.096 (2.438)	.118 (2.997)	.132 (3.353)

NASM35275 PIN 1/	J Slot width min	J Slot width max	T Slot depth min	T Slot depth max	E drill location nom	Drilled hole size nom	R radius max	Tensile strength LB-min (KN-min)
MS35275-215	.031 (.787)	.039 (.991)	.030 (.762)	.038 (.965)	.035 (.889)	.037 (.940)	.017 (.432)	.480 (2.151)
MS35275-229	.039 (.991)	.048 (1.219)	.035 (.889)	.045 (1.143)	.043 (1.092)	.046 (1.168)	.021 (.533)	.730 (3.272)

1/ National Aerospace Standard NASM35275. Not to be ordered with this slash sheet (see note 1).

To order screws individually use the following NASM PINs:

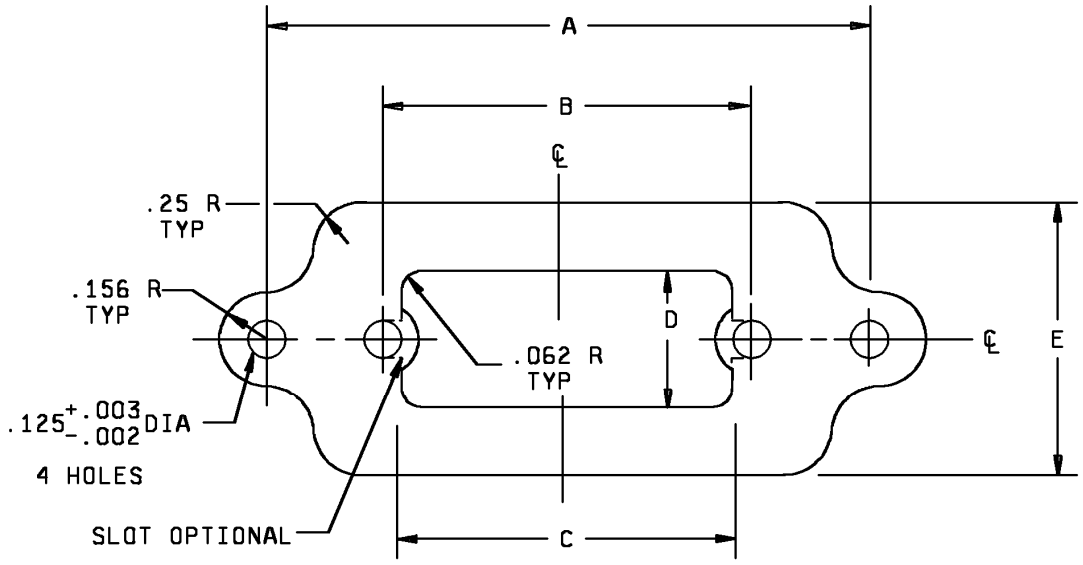
a. #4 screw MS35275-215
b. #6 screw MS35275-229

NOTES:

1. Screws are shown for reference purposes only.
2. Material corrosion resisting, in accordance with SAE-AMS-STD-66, AISI 302, 303, 304, 305, 316, 384 (developed for cold heading) or type UNS S30430 in accordance with the chemical requirements of ASTM A493.
3. Treatment: Passivation shall be in accordance with SAE-AMS-QQ-P-35.
4. Magnetic permeability: These screws shall have a magnetic permeability of 2.0 μ maximum (air = 1.0μ) for a field strength of H = 200 oersteds, using a magnetic indicator in accordance with ASTM A342.
5. Threads: Threads shall be in accordance with FED-STD-H28.
6. Threads: Screws shall have complete threads extending to within two threads of the head, or closer if practicable.
7. Drilled hole is to be approximately at right angles to the slot and may be permitted to break through the bottom of the slot. All sharp edges shall be removed.
8. Copies of NASM35275 are available online at http://www.aia-aerospace.org/, or may be purchased from Aerospace Industries Associations of America Inc., 1250 Eye Street N.W., Washington, DC 20005-3924.

FIGURE 15. Slotted screw.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



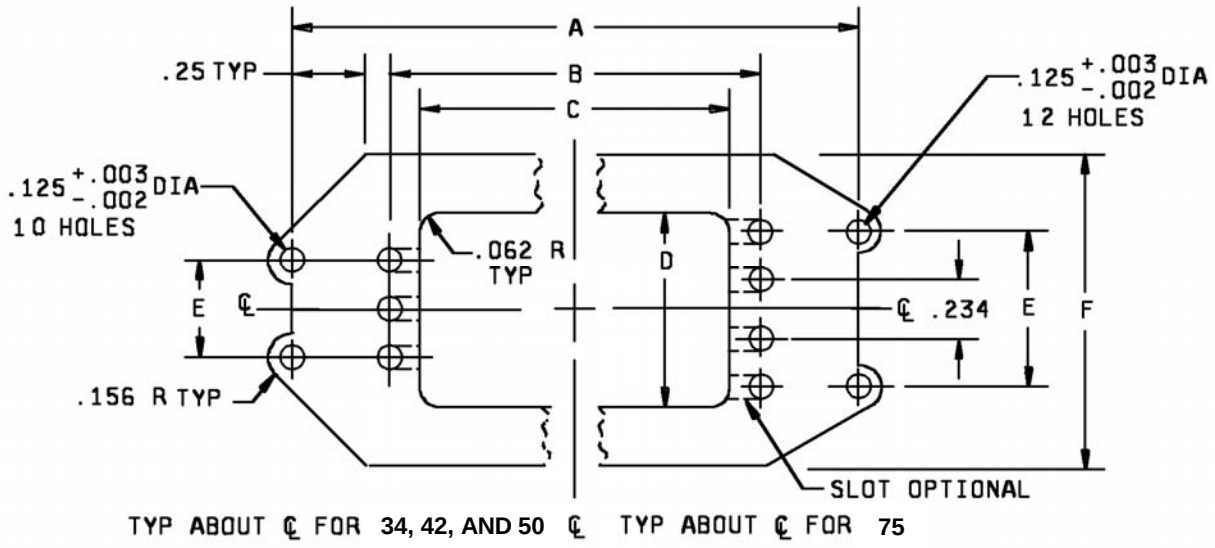
Inches	mm
.002	0.05
.003	0.08
.062	1.57
.125	3.18
.156	3.96
.25	6.35

Retaining plate designator	Connector size (number of contacts)	A ±.005 (0.13 mm)	B ±.005 (0.13 mm)	C (mm)	D (mm)	E (mm)
M18198-1	9	2.016 (51.21)	1.000 (2.54)	.875 (22.23)	.406 (10.31)	.953 (24.21)
M18198-2	14	1.953 (49.61)	.937 (23.80)	.812 (20.62)	.469 (11.91)	1.016 (25.81)
M18198-3	20	2.266 (57.56)	1.250 (31.75)	1.125 (28.58)	.469 (11.91)	1.016 (25.81)
M18198-4	26	2.328 (59.13)	1.312 (33.32)	1.187 (30.15)	.593 (15.06)	1.141 (28.98)

See notes at end of figure.

FIGURE 16. Retaining plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



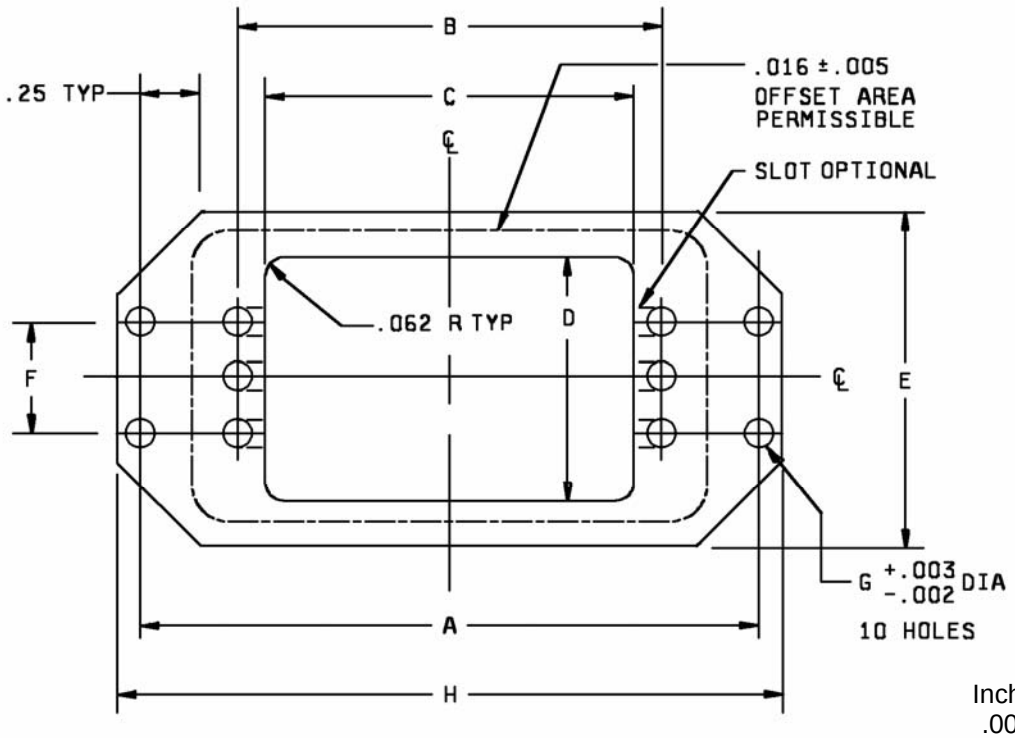
Inches	mm
.002	0.05
.003	0.08
.062	1.57
.125	3.18
.156	3.96
.234	5.94
.25	6.35

Retaining plate designator	Connector size (number of contacts)	A ±.005 (0.13 mm)	B ±.005 (0.13 mm)	C (mm)	D (mm)	E ±.005 (0.13 mm)	F (mm)
M18199-1	34	2.703 (68.66)	1.687 (42.85)	1.438 (36.53)	.750 (19.05)	.468 (11.89)	1.328 (33.73)
M18199-2	42	3.016 (78.89)	2.000 (50.80)	1.750 (44.45)	.750 (19.05)	.468 (11.89)	1.328 (33.73)
M18199-3	50	3.422 (86.92)	2.281 (57.94)	2.031 (51.60)	.750 (19.05)	.468 (11.89)	1.453 (36.91)
M18199-4	75	3.422 (86.92)	2.281 (57.94)	2.031 (51.60)	1.109 (28.17)	.766 (19.46)	1.797 (45.64)

See notes at end of figure.

FIGURE 16. Retaining plate - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.016	0.41
.062	1.57
.25	6.35

Retaining plate designator	Connector size (number of contacts)	A ±.005 (0.13 mm)	B ±.005 (0.13 mm)	C (mm)	D (mm)	E (mm)	F ±.005 (0.13 mm)	G (mm)	H (mm)
M18200-1	66	2.984 (75.79)	1.969 (50.01)	1.718 (43.64)	1.125 (28.58)	1.688 (42.88)	.500 (17.70)	.125 (3.18)	3.296 (83.72)
M18200-2	104	3.500 (83.90)	2.375 (60.33)	2.125 (53.98)	1.486 (37.74)	2.000 (50.80)	.875 (22.23)	.150 (3.81)	3.875 (98.43)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with SAE-AMS-QQ-A-250/8, temper H32, .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.

FIGURE 16. Retaining plate - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figures 1 through 16.

Connector parts described herein are intended for use with MIL-DTL-28748/1 through MIL-DTL-28748/6.

NASM screws and nuts are not ordered on this slash sheet, they are shown for reference only.

PIN: As shown in the following example.

M28748/17	M18204-1F
Basic number of specification	designator from figures

For supersession information see table I.

TABLE I. Superseded MS sheets.

MS sheet	Dated
MS18192	1 June 1983
MS18193	26 April 1990
MS18194	11 April 1974
MS18195	20 May 1970
MS18196	23 June 1972
MS18197	23 June 1972
MS18198	23 June 1972
MS18199	23 June 1972
MS18200	20 May 1970
MS18203	26 April 1990
MS18204	26 April 1990
MS18205	23 June 1972
MS24132	1 June 1976
MS24133	3 January 1979

Reference documents. In addition to MIL-DTL-28748, this document references the following:

MIL-A-8625	FED-STD-H28	SAE-AMS-QQ-A-250/8
MIL-DTL-28748/1	ASTM A342	SAE-AMS-QQ-A-250/11
MIL-DTL-28748/2	ASTM A493	SAE-AMS-QQ-P-35
MIL-DTL-28748/3	ASTM A532	SAE-AMS-QQ-S-763
MIL-DTL-28748/4	ASTM B582	SAE-AMS-STD-66
MIL-DTL-28748/5	NASM21044	
MIL-DTL-28748/6	NASM35275	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5935-4572-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:19Z
Check the source to verify that this is the current version before use.